

# Analysis of Renewable Energy Project Delays on The Indonesia's Energy Transition Process, Case Study: Java Bali System

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**Abstract**— In order to reduce greenhouse gas emissions, all countries, including Indonesia, are trying to utilize clean energy in their energy supply. To be able to realize the use of clean energy and reduce the effect of greenhouse gases, Indonesia has decided to accelerate the energy transition so that it can achieve NZE by 2060. The problem that is often encountered in power generation projects is delays in power generation projects, which have not been considered in the current power plant planning. Based on this, this paper will conduct a study by considering aspects of the success and delay ratio of a power plant project in planning. By considering these aspects, plant planning will be carried out using the MILP method and then the influence of the delay and success ratio will be analyzed on the renewable energy mix, LCOE, and the plants that will replace it. Based on the results obtained, The extent of the delay in capacity refers to the success ratio, delays, and achievements of the power plant projects. Delayed VRE PP up to 2 GW, and based on the results, it was found that these generators would be substituted with Gas Machines up to 2000 MW. This affects the LCOE of the system, increasing it from 9.77 c\$/kWh to up to 8 c\$/kWh. This LCOE calculation does not yet account for the unavailability of LNG, which would lead the Gas Machine to utilize more expensive HSD (High-Speed Diesel), potentially causing a further increase in the LCOE.

**Keywords**— Renewable energy, generation expansion planning, energy transition, project delays, LCOE

## I. INTRODUCTION

Currently, the issue of global warming is a matter of concern to all countries in the world, including Indonesia. Various efforts have been made to reduce CO<sub>2</sub> emissions and greenhouse gases (GHG). Indonesia is committed to implementing the Paris Agreement through [1].

Indonesia has made several efforts, one of them is accelerating the energy transition, especially in the electricity sector. Based on [2], Indonesia is targeting an increase in its renewable energy mix of 23% in 2025 and 31% in 2050. The Indonesian government's efforts to increase the utilization of clean energy. Based on [3], renewable energy generation capacity has reached 12.72%.

Generation expansion planning with consider the renewable energy has been done in various country, such as USA [4], Ireland [5], France [6], China [7]–[9], Brazil [10], Portugal [11], Egypt [12], Pakistan [13], Oman [14], South Africa [15], Bangladesh [16], Malaysia [17] and Indonesia [18]–[22] have considered the environmentally friendly resource or RES in their GEP model. Based on these studies, considering VRE can lead to increasing or decreasing the total generation cost.

A statistical residual load duration curve (S-RLDC), a technique to simplify the duration curve load method, was used to model the load for GEP in China [9]. The S-RLDC modelled the generation of renewable energy as a negative load. In [10], a GEP using multiple objective functions

considering renewable energy was successfully conducted. In the study, the use of non-hydro renewable energy in the Brazilian power system could be increased. Another study included external costs of environmental impacts on GEP in China to accommodate the entry of intermittent power plants [7].

In addition, several studies on GEP have considered Indonesia's high share of RES, including dispatchable and variables RES (VREs). For example, research [22] incorporates high share VREs of wind in GEP, while research [23] considers the emission factor among the VREs factor. Moreover, the potential of local energy sources has also been studied in [24], which analyzes utilizing biomass in eastern Indonesia. Furthermore, research [25] modelled a GEP on isolated systems, considering local energy sources and the interconnections between systems.

However, based on historical data from [3], the success ratio in a power plant project is not always as expected. Based on the data that has been processed, the success rate of the hydro power plant (hydro PP) project is 75%, mini hydro power plant (mini hydro PP) 2x%, and PV power plant (PP) 0% for the Java Bali system until 2021. Delays and success of a project can be caused by several factors, such as legal aspects, land acquisition, power purchase agreements, or even failures in the construction process.

Currently, plant planning taking into account aspects of success or delay in Indonesia has not been carried out. In planning [18], [20], [22], [23], [26], [27], the GEP still does not consider these aspects. Because of this, a study is needed that considers the ratio of success and also delays in plant planning.

In this paper, we will discuss the influence of delay and success ratios on plant planning. The contribution of this paper is that it can provide an overview of the impacts and mitigation that need to be carried out if power plant development planning takes these two aspects into consideration.

The rest of the paper is as follows: Section II will discuss the objective function and constraints in the GEP. Section III describes the methodology of the research to be carried out. Section IV discusses the results of the research. Finally, section V presents the conclusions from the results of this research.

## II. GENERATION EXPANSION PLANNING

### A. Objective Function

In this paper, consider the objective function as in (1). This aims to obtain the optimal combination of generators to meet the supply of electricity by considering several aspects such as delays. The objective function is to minimize the discounted generation costs while adhering to certain constraints [21]–[23], [25]. The calculation of GEP is based on the total fixed cost, variable operating and maintenance cost, and fuel cost. The total fixed cost consists of the total annualized investment and fixed operating and maintenance costs as in (2)–(10).

$$\min \text{Cost} = C_{\text{Fix}}^{\text{total}} + C_{\text{VO\&M}}^{\text{total}} + C_{\text{Fuel}}^{\text{total}} \quad (1)$$

$$C_{\text{Fix}}^{\text{total}} = C_{\text{Annual}}^{\text{total}} + C_{\text{FO\&M}}^{\text{total}} \quad (2)$$

$$C_{\text{Annual}}^{\text{total}} = C_{\text{Inv}}^{\text{total}} \times \frac{(1-\text{Tax}) \times \frac{\text{NAF}}{\text{EcoLife}}}{\text{RAF}} \quad (3)$$

$$C_{\text{Inv}}^{\text{total}} = \sum_y \sum_g DF_y \times (C_{\text{inv}}^g \times 1000 \times P_{\text{max}}^g \times NB_g^y) \quad (4)$$

$$C_{\text{FO\&M}}^{\text{total}} = \sum_y \sum_g DF_y \times [C_{\text{FO\&M}}^g \times P_{\text{max}}^g (N_g + \sum_{i \leq y} NB_g^i)] \quad (5)$$

$$C_{\text{VO\&M}}^{\text{total}} = \sum_t \sum_g DF_{t \in y} \times L_t \times (C_{\text{VO\&M}}^g \times G_g^t) \quad (6)$$

$$C_{\text{Fuel}}^{\text{total}} = \sum_t \sum_g DF_{t \in y} \times L_t \times (\text{Heat Rate} \times C_{\text{fuel}}^g \times G_g^t) \quad (7)$$

$$DF = \frac{1}{(1+D)^y} \quad (8)$$

$$\text{NAF} = \frac{1 - (1 + \text{WACC} + \text{Infation rate})^{-\text{Ecolife}}}{\text{WACC} + \text{Infation rate}} \quad (9)$$

$$\text{RAF} = \frac{1 - (1 + \text{WACC})^{-\text{Ecolife}}}{\text{WACC}} \quad (10)$$

With,

- $C_{\text{Fix}}^{\text{total}}$  : Total fixed cost (\$/kW/yr)
- $C_{\text{Annual}}^{\text{total}}$  : Total annualized build cost (\$/kW/yr)
- $C_{\text{Inv}}^{\text{total}}$  : Total build cost (\$/kW)
- $C_{\text{FO\&M}}^{\text{total}}$  : Total fixed O&M cost (\$/kW/yr)
- $C_{\text{VO\&M}}^{\text{total}}$  : Total var O&M cost (\$/MWh/yr)
- $C_{\text{Fuel}}^{\text{total}}$  : Total fuel cost (\$/MWh)
- $DF_y$  : Discount factor
- $D$  : Discounted Rate (%)
- $P_{\text{max}}^g$  : Generator max capacity (MW)
- $NB_g^y$  : Number of generator
- $G_g^t$  : Generation of generator  $g$  in  $t$  period (MWh)
- $C_{\text{inv}}^g$  : Build cost of generator  $g$  (\$/kW)
- $C_{\text{FO\&M}}^g$  : FO&M cost generator  $g$  (\$/kW/yr)
- $C_{\text{VO\&M}}^g$  : VO&M cost generator  $g$  (\$/MWh/yr)
- $C_{\text{fuel}}^g$  : Fuel cost generator  $g$  (\$/MWh/yr)

### B. Constraints

During the optimization process, several constraints are taken into consideration. The power balance constraint ensures that the total generation during a specific period must meet or exceed the load demand at that time, as depicted in equation (11). Additionally, the total generation is limited by the available installed generating capacity, as stated in equation (12). The number of new power plants constructed cannot surpass the maximum potential, as outlined in equation (13). However, the total installed capacity must be adequate to meet the yearly peak load plus a reserve margin specified in equation (14).

$$\sum_{g=1}^{NG} G_g^t \geq D^t, \forall t \quad (11)$$

$$G_g^t \leq P_{\text{max}_g} \left( N_g + \sum_{i \leq y, y=1}^{NY} NB_g^i \right), \forall g, t \quad (12)$$

$$\sum_{i \leq y, y=1}^{NY} NB_g^i \leq NB_{\text{max}_g}^i, \forall g \quad (13)$$

$$\sum_{g=1}^{NG} P_{\text{max}_g} \left( N_g + \sum_{i \leq y, y=1}^{NY} NB_g^i \right) \geq PL^y (1 + RM), \forall y \quad (14)$$

With,

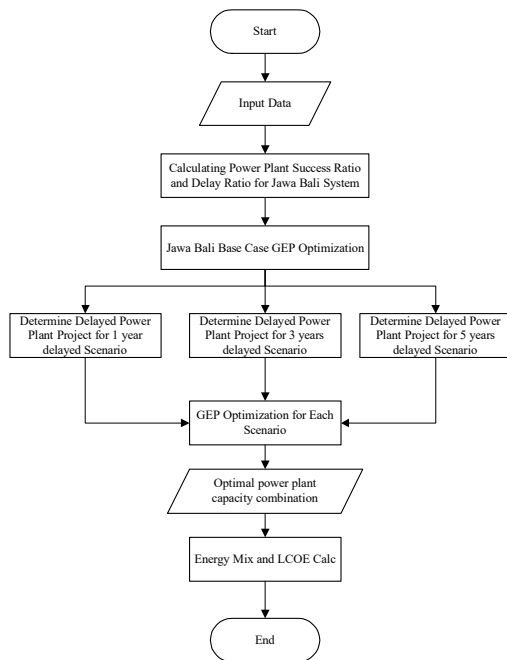


Fig. 1. The Generation Expansion Planning Workflow

$D^t$  : Demand in t period (MWh)  
 $NB_{max}^i$  : Max Number of generator g that can be built  
 $PL^y$  : Peak load (MW)  
 $RM$  : Reserve margin (%)

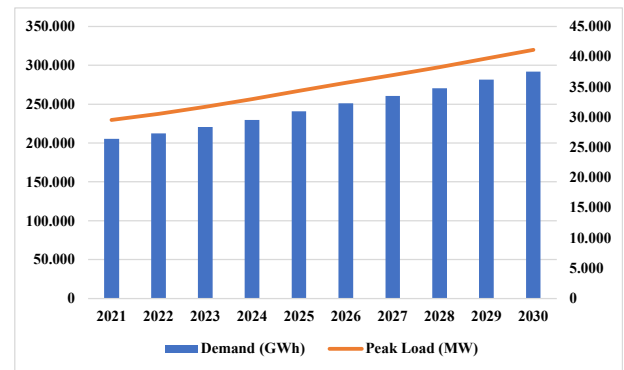


Fig. 2. Java Bali System Demand and Peak Load

### III. METHODOLOGY

The process of the research is illustrated in Fig 1. Information required for this research includes predicted energy demand and peak load for the planning period, potential power plant and primary energy resources. Annual energy demand and peak load as illustrated in Fig 2. The information on the potential power plant options includes plant size, heat rate, maintenance rate, forced outage rate, lifetime, investment cost, and operating costs, which are presented in Table I [28], [29].

The study in this paper uses the Java Bali system as a system test, which is the largest electricity system in Indonesia. In this paper, a comparison of GEP results from normal conditions with a delay scenario is carried out. The

TABLE I. POWER PLANT CANDIDATE DATA [28],[29]

| Type       | Heat Rate (GJ/MWh) | Maintenance Rate (%) | FOR (%) | Technical Lifetime (years) | Investment Cost (\$/kWe) | Fixed O&M Cost (\$/kWe/y) | Variable O&M Cost (\$/MWh) |
|------------|--------------------|----------------------|---------|----------------------------|--------------------------|---------------------------|----------------------------|
| Gas        | 8.37               | 2.88                 | 3       | 25                         | 800                      | 18                        | 1                          |
| Hydro      | -                  | 11.51                | 4       | 50                         | 2,000                    | 6.6                       | 1                          |
| Biomass    | 9                  | 11.51                | 7       | 25                         | 1,700                    | 47.6                      | 3                          |
| Wind       | -                  | 0.31                 | 3       | 27                         | 1,500                    | 39.55                     | 0.8                        |
| Solar      | -                  | 5                    | 5       | 25                         | 790                      | 24.7                      | 0.4                        |
| Coal       | 9                  | 11.51                | 7       | 30                         | 1,650                    | 45.3                      | 0.13                       |
| Geothermal | 4.32               | 7.67                 | 10      | 30                         | 4,000                    | 65                        | 0.37                       |

TABLE II. EXISTING POWER PLANT DATA (IN MW)

| Existing Gen  | 2021   | 2022   | 2023   | 2024   | 2025   | 2026   | 2027   | 2028   | 2029   | 2030   |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Hydro PP      | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  | 2,615  |
| Mini Hdro PP  | 150    | 150    | 150    | 150    | 150    | 150    | 150    | 150    | 150    | 150    |
| Geothermal PP | 1225   | 1225   | 1225   | 1225   | 1225   | 1225   | 1225   | 1225   | 1225   | 1225   |
| Gas PP        | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  | 1,467  |
| CCGT          | 9,477  | 8,014  | 6,929  | 6,929  | 6,929  | 6,929  | 6,929  | 6,929  | 6,929  | 6,929  |
| Gas Machine   | 182    | 182    | 182    | 182    | 182    | 182    | 182    | 182    | 182    | 182    |
| Diesel        | 152    | 152    | 152    | 152    | 152    | 152    | 152    | 152    | 152    | 152    |
| Coal          | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 | 22,479 |
| Coal with HSD | 927    | 927    | 927    | 927    | 927    | 927    | 927    | 927    | 927    | 0      |
| Biomass       | 5      | 5      | 5      | 5      | 5      | 5      | 5      | 5      | 5      | 5      |
| Waste         | 15     | 15     | 15     | 15     | 15     | 15     | 15     | 15     | 15     | 15     |
| Excess        | 3      | 3      | 3      | 3      | 3      | 3      | 3      | 3      | 3      | 3      |

TABLE III. POWER PLANT CANDIDATE

|               | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|---------------|------|------|------|------|------|------|------|------|------|------|
| Hydro PP      | 110  | 0    | 0    | 0    | 0    | 50   | 0    | 0    | 0    | 0    |
| Coal PP       | 4215 | 924  | 0    | 0    | 1660 | 1660 | 0    | 0    | 0    | 0    |
| Geothermal PP | 0    | 0    | 130  | 65   | 330  | 265  | 55   | 190  | 75   | 805  |
| PV PP         | 0    | 145  | 235  | 550  | 1240 | 110  | 140  | 140  | 140  | 140  |
| Waste         | 9    | 5    | 0    | 35   | 183  | 0    | 0    | 0    | 0    | 0    |
| Mini Hydro    | 45   | 38   | 144  | 102  | 76   | 12   | 0    | 0    | 0    | 0    |
| Wind PP       | 0    | 0    | 0    | 160  | 100  | 0    | 0    | 0    | 0    | 0    |
| PS            | 0    | 0    | 0    | 0    | 1040 | 0    | 0    | 943  | 760  | 1000 |
| CCGT          | 2110 | 1279 | 200  | 0    | 100  | 0    | 0    | 0    | 0    | 0    |

TABLE IV. DELAY AND SUCCESS RATIO

| Hydro PP   |     |     | Wind PP |     |      | Biomass                                                                     |     |      |
|------------|-----|-----|---------|-----|------|-----------------------------------------------------------------------------|-----|------|
| %SR        | %DR | %F  | %SR     | %DR | %F   | %SR                                                                         | %DR | %F   |
| 75%        | 33% | 25% | 0%      | 0%  | 100% | 0%                                                                          | 0%  | 100% |
| Geothermal |     |     | PV PP   |     |      | %SR : Success Ratio<br>%DR : Delay Ratio<br>%F : Project Failure Percentage |     |      |
| %SR        | %DR | %F  | %SR     | %DR | %F   |                                                                             |     |      |
| 19%        | 33% | 81% | 0%      | 0%  | 100% |                                                                             |     |      |

delay scenario is divided into 3 scenarios, namely 1 year delay, 3 years, and 5 years. The delay is applied to renewable energy plants that have a failure ratio, such as PV PP, Wind PP, and Geothermal PP. Existing generating capacity data can be seen in Table II and the planning power plant in Table III. Success ratio and delay ratio data can be seen in Table IV.

#### IV. RESULTS AND DISCUSSION

The addition of power plant capacity, both VRE PP and conventional PP, has an impact on energy supply in the Java Bali System. This section will present and discuss the impact of VRE PP delays on the composition of power plant capacity, energy mix, and LCOE.

##### A. Impact of Power Plant Project Delay on Capacity Composition

Based on research that has been done, taking into account the delays in PV PP 550 MW generators in 2024 and 1240

MW in 2025, Geothermal PP 330 MW in 2025 and 265 in 2026, as well as wind PP 160 MW in 2024 and 100 MW in 2025, resulting in changes to the power plants composition in the Java Bali System.

In the base case scenario, the total generating capacity in the Java Bali system in 2030 is 60.41 GW. By considering the delay aspect of the power plant construction project, the result is that the system will build additional power plants to be able to substitute for the power plants shortfall. In this research, the power plants that can replace the shortage of power plant capacity is assumed to be a gas generator or gas machine. The choice of this type of generator considers relatively fast supply, both conventional and Mobile Power Plant (MPP). The results of plant planning for the base case

TABLE V. POWER PLANT TOTAL CAPACITY

|      | Base Case | Scenario 1-year Delayed | Scenario 3-year Delayed | Scenario 5-year Delayed |
|------|-----------|-------------------------|-------------------------|-------------------------|
| 2021 | 45,186.00 | 45,186.00               | 45,186.00               | 45,186.00               |
| 2022 | 47,577.00 | 47,577.00               | 47,577.00               | 47,577.00               |
| 2023 | 48,286.00 | 48,286.00               | 48,286.00               | 48,286.00               |
| 2024 | 49,197.90 | 48,487.90               | 48,487.90               | 48,487.90               |
| 2025 | 53,926.90 | 52,966.90               | 52,256.90               | 52,256.90               |
| 2026 | 56,164.00 | 57,569.00               | 55,899.00               | 55,899.00               |
| 2027 | 56,218.90 | 56,483.90               | 56,928.90               | 56,218.90               |
| 2028 | 57,491.90 | 57,491.90               | 59,161.90               | 57,491.90               |
| 2029 | 58,466.90 | 59,066.90               | 59,331.90               | 59,776.90               |
| 2030 | 60,411.80 | 61,311.80               | 61,611.80               | 63,281.80               |

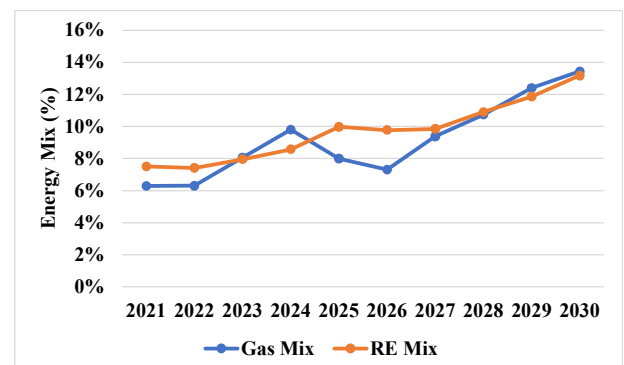


Fig. 3. Java Bali System Gas Mix Vs RE Mix

TABLE VI. JAVA BALI SYSTEM LCOE

|                | Base Case | Scenario 1-year Delayed | Scenario 3-year Delayed | Scenario 5-year Delayed |
|----------------|-----------|-------------------------|-------------------------|-------------------------|
| LCOE (c\$/kWh) | 7.7       | 7.9                     | 8                       | 8,1                     |

scenario and delay scenario will produce a plant composition as in Table V.

To anticipate the delay in the construction of RE power plants, as indicated by the research findings, there is a need for the substitution of power plants capacity. The results show the establishment of up to 1000 MW gas machines in the scenario of a 1-year delay and up to 1800 MW gas machine in the scenarios of 3 and 5-year delays. These 1000 MW and 1800 MW gas machines are intended to accommodate the delay in RE power plants amounting to 2645 MW, including PV PP, Geothermal PP, and Wind PP. The delay in these power plants results in an increase in the installed capacity in the Java Bali system from 60.41 GW to up to 63.28 GW.

### B. Energy Mix and LCOE

Based on the installed capacity, changes in the energy mix in the Java Bali system are obtained. This change primarily occurs in the RE mix, which experienced construction delays, resulting in an increase in the gas energy mix. The results indicate a decrease in the RE energy mix when PV PP delayed year in 2024, as shown in Fig. 3. This can occur due to the delayed PV PP up to 1810 MW.

With changes in power plants capacity and energy mix, the Java Bali electrical system experiences a shift in the LCOE of generation. As shown in Table VI, the LCOE will increase from the base case of 7.7 c\$/kWh to 8.1 c\$/kWh in the delay scenario. This LCOE does not consider the unavailability of LNG, which would lead the Gas Machine to use more expensive HSD (High-Speed Diesel), potentially causing a further increase in the LCOE.

### V. CONCLUSION

Based on the research findings, it can be determined that delays in power plants project, such as RE PP, can influence the capacity planning and the LCOE of the system. This study focuses on the delays in RE such as PV PP, Geothermal PP, and Wind PP. The extent of the delay in capacity refers to the success ratio, delays, and achievements of the power plant projects. Delayed VRE PP up to 2 GW, and based on the results, it was found that these generators would be substituted with Gas Machines up to 2000 MW. This affects the LCOE of the system, increasing it from 7.7 c\$/kWh to up to 8.1 c\$/kWh. This LCOE calculation does not yet account for the unavailability of LNG, which would lead the Gas Machine to utilize more expensive HSD (High-Speed Diesel), potentially causing a further increase in the LCOE.

Moreover, the impact of these delays appears to be relatively small, due to the limited utilization of VRE. This situation might differ if delays occurred in base load power plants such as Coal PP, Hydro PP, and Biomass PP. Therefore, further studies are needed to assess the impact of base load generators on the energy mix, LCOE, and emissions.

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